P1 — Hierarchical Concept-Based Explainable-by-Design Models

Explainable and Trustworthy AI Course

Politecnico di Torino - 2025/2026

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Project. The project aims to design and evaluate concept-based explainable-by-design models in which concepts are explicitly organized in a hierarchy, enabling explanations and predictions at multiple levels of abstraction.

Overview.

Concept-based explainable models aim to improve interpretability by expressing predictions in terms of human-understandable concepts [4]. In most existing approaches, however, concepts are represented as a flat set of human-defined symbols. This simplification overlooks that concepts are naturally organized in hierarchies, from general to more specific ones. For example, specific concepts such as *beak* can be organized into higher-level parts such as *head*.

Hierarchical structures over concepts can be obtained in different ways. For symbolic concepts, hierarchies can be derived from existing ontologies, such as WordNet, which define relationships between concepts and organize specific concepts into more general ones. Alternatively, hierarchical structures can be automatically constructed, for instance by querying a large language model, extending recent approaches that generate flat concept sets by introducing relations among concepts.

Recent works have started to explore hierarchical structures in concept-based models and interpretable architectures, showing that multi-level representations can improve interpretability and predictive performance [2, 3, 5]. Moreover, hierarchical reasoning enables explanations at different levels of granularity and supports a more nuanced understanding of model behavior. For example, a model may correctly identify a high-level category even when it fails at a fine-grained classification. In parallel, recent research has highlighted that not all classification errors are equally severe: confusing semantically similar classes is less problematic than confusing unrelated ones [1]. Hierarchical relationships between classes can be used to quantify the severity of mistakes and design more informative evaluation strategies. Despite these advances, current concept-based explainable models rarely integrate hierarchical structures explicitly, and there is still a lack of unified frameworks that combine hierarchical concept representations, interpretable predictions, and hierarchy-aware evaluation.

Goal.

The goal of this project is to investigate hierarchical concept-based explainable models by designing a framework in which concepts are structured across multiple levels of abstraction. The project aims to assess whether hierarchical representations improve interpretability, enable richer explanations, and allow for a more informative analysis of model predictions and errors.

Required analysis, implementation, and evaluation.

- **Literature Review.** Conduct a systematic review of concept-based explainable models, with particular focus on concept bottleneck models, coarse-to-fine concept discovery, and hierarchical prototype-based methods. Review also hierarchy-aware evaluation methods for classification errors and semantic mistake severity.
- **Identification of Research Gaps.** Identify limitations of current concept-based approaches, such as flat concept representations, insufficient modeling of coarse-to-fine semantic relations, and limited evaluation of mistakes at different abstraction levels.
- **Implementation.** Design and implement a hierarchical concept-based model. Possible directions include:
  - defining a multi-level concept bottleneck architecture, e.g., by extending an existing CBM with *is-a* concept relations;
  - constraining the prediction process so that coarse concepts support fine-grained concepts;
  - generating explanations at multiple levels of granularity.
- **Evaluation.** Evaluate the proposed model on at least one dataset using both predictive and interpretability metrics. The evaluation should include standard predictive performance, concept quality, and hierarchy-aware analysis, for example, by assessing whether the model makes semantically better mistakes or whether explanations become more meaningful across different levels of abstraction.

References

[1] Luca Bertinetto et al. "Making better mistakes: Leveraging class hierarchies with deep networks". In: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 2020, pp. 12506–12515.

[2] Peter Hase et al. "Interpretable image recognition with hierarchical prototypes". In: *Proceedings of the AAAI conference on human computation and crowdsourcing*. Vol. 7. 2019, pp. 32–40.

[3] Federico Pittino, Vesna Dimitrievska, and Rudolf Heer. "Hierarchical concept bottleneck models for vision and their application to explainable fine classification and tracking". In: *Engineering Applications of Artificial Intelligence* 118 (2023), p. 105674.

[4] Eleonora Poeta et al. "Concept-based Explainable Artificial Intelligence: A Survey". In: *ACM Comput. Surv.* (Nov. 2025). ISSN: 0360-0300. DOI: 10.1145/3774643. URL: https://doi.org/10.1145/3774643.

[5] Ao Sun et al. "Boosting Concept Bottleneck Models with Supervised, Hierarchical Concept Learning". In: *Paper under review.* *https://openreview.net/forum?id=Q9ZOc1Rb5i* ().